

RESULTS

Results

01 · INSTRUMENTAL VARIABLES (IV / 2SLS)

effect with an endogenous predictor

Table 1. First stage (instrument strength)

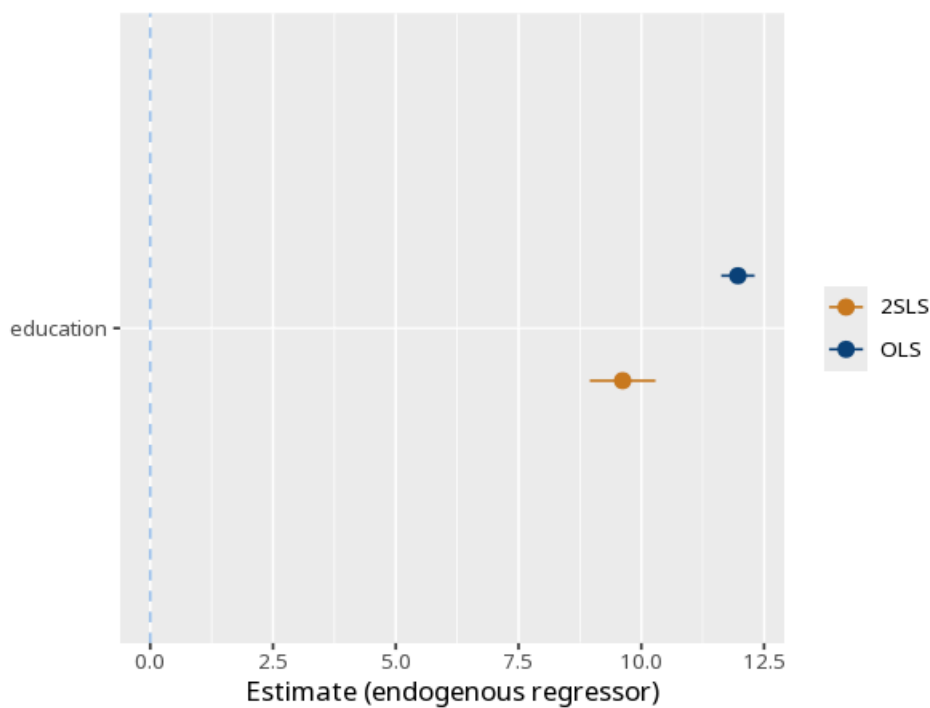
Endogenous	Instrument	Coef.	SE	Partial F	p
education	mother_education	0.54	0.02	507.83	<.001

Table 2. 2SLS coefficients

	OLS	2SLS
(Intercept)	207.67 (2.49) [202.77, 212.57]	240.20 (4.71) [230.92, 249.49]
education	11.97 (0.17) [11.63, 12.31]	9.62 (0.34) [8.95, 10.29]
experience	0.62 (0.06) [0.51, 0.72]	0.63 (0.07) [0.50, 0.76]
Num.Obs.		250
RMSE		7.52
F		447.72
Weak-instrument F = 507.83, p < .001 (rule of thumb: F > 10)		
Wu-Hausman = 1029.75, p < .001		
Sargan: — (just-identified)		

diagnostics: weak-instrument (first-stage F), Wu-Hausman endogeneity, and Sargan over-identification (when applicable).

Figure. Coefficients



Understanding the numbers

Endogenous. Which endogenous regressor this first-stage row predicts. Here, this row predicts education.

Instrument. Which instrument this first-stage row reports on. Here, this row is for the instrument mother_education.

Coef.. The first-stage coefficient: how strongly this instrument predicts the endogenous regressor. Here, the first-stage coefficient is 0.54.

SE. The standard error of that first-stage coefficient. Here, SE = 0.02.

Partial F. The partial F-statistic for this instrument - the rule of thumb is $F > 10$ for a strong instrument. Here, partial F = 507.83.

p. The significance of the first-stage relationship between this instrument and the endogenous regressor. Here, $p < .001$.

OLS. The naive OLS estimate for the endogenous predictor, ignoring its possible endogeneity - shown for comparison. Here, OLS gives 11.97.

2SLS. The 2SLS estimate for the endogenous predictor - read as the causal effect only if the instrument is valid (relevant, and affecting the outcome only through the endogenous predictor). Here, 2SLS gives 9.62.

Num.Obs.. The number of observations the 2SLS model was fit on. Here, $N = 250$.

RMSE. The typical size of the 2SLS model's prediction error, in the outcome's own units. Here, RMSE = 7.52.

F. The overall (structural) F-test for the second-stage 2SLS model. Here, $F = 447.72$.

How to read this test

Check the first-stage F first (rule of thumb > 10 = strong instruments). Then read the 2SLS B for the endogenous predictor as the causal effect, with p/CI. This causal reading rests on the instrument being valid - relevant and affecting the outcome only through the endogenous predictor (the exclusion restriction). That is an assumption you must justify on theory; the Sargan test only checks over-identification (extra instruments), not validity, and is unavailable for just-identified models. Method: R ivreg / AER package; weak-instrument F (Stock & Yogo, 2005). Your significance threshold (α) is 0.05.

APA template: The 2SLS estimate for education was $B=9.62$, 95% CI [8.95, 10.29], $p < .001$ (first-stage $F=507.83$).

Statistical basis: Two-stage least squares (2SLS) estimation (Fox J, Kleiber C, Zeileis A (2024). ivreg: Instrumental-Variables Regression by 2SLS, 2SM, or 2SMM, with Diagnostics. R package.); Weak-instrument first-stage F rule of thumb (Stock, J. H., Yogo, M. (2005). "Testing for Weak Instruments in Linear IV Regression." In Identification and Inference for Econometric Models (Andrews & Stock, eds.), Cambridge University Press, 80-108.); Heteroscedasticity-robust standard errors (Zeileis A, Köll S, Graham N (2020). "Various Versatile Variances: An Object-Oriented Implementation of Clustered Covariances in R." Journal of Statistical Software, 95(1), 1-36.); Wu-Hausman endogeneity test (Wu, D. M. (1973). "Alternative tests of independence between stochastic regressors and disturbances." Econometrica, 41(4), 733-750.); Sargan over-identification test (Sargan, J. D. (1958). "The estimation of economic relationships using instrumental variables." Econometrica, 26(3), 393-415.). Full references in the exported CITATIONS.txt.

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